

Alexandre Labi DAHOUE

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Profil :

I am a senior studying engineering with a focus on numerical modeling and simulation. I have acquired a diverse set of skills—both technical and interpersonal—during my years of study, and I am extremely determined to apply them to innovative endeavors and large-scale projects. I am very detail-oriented, resourceful, and curious, with a keen appreciation for art, science, beauty, and aesthetics. I have a deep commitment to teamwork and progress; I have an in-depth understanding of numerical optimization, stochastic simulation, statistics, probability, data analysis, Markov chains, statistical inference, operations research, neural networks, and numerical modeling. I am motivated to learn, acquire new skills, and excel in cutting-edge technologies.

Education :

2023-2026 : Bachelor of Science in Mathematical Engineering and Modeling

2021-2023 : Preparatory Courses for Engineering Schools

2020-2021 : High School Diploma

Research Projects:

AquaTwin-Drip v1.0 Hybrid Digital Twin for Controlling Drip Irrigation : Development of a descriptive digital twin (AquaCrop FAO physical model and Richards' equation) and a predictive digital twin (Data-Driven AI engine) for drip irrigation scheduling; engineering thesis; in progress. [Alex542277/AquaTwin-Drip-Hybrid-Digital-Twin-for-Controlling-Drip-Irrigation: The Challenge of Irrigation: From Empirical Decision-Making to Precision Management.](https://www.linkedin.com/posts/alex-dahoue-b7411536a_certificate-activity-7405674191913652224-0BsV?utm_source=share&utm_medium=member_desktop&rcm=ACoAAft_ZUGBtQ-1PtrzQAvTY0laB0NQ2GM0I4)

Fractional-Order Modeling of Saturation Dynamics and Production Optimization in Black-Oil Reservoir Simulation, Elsevier preprint, currently being published in *Sciences des Structures et de la Matière*, African and Malagasy Council for Higher Education (CAMES)

Dynamics of Leptospirosis, Accepted and presented on December 1, 2025, at the Second International Conference of the Pan-African University (PAU INTERNATIONAL CONFERENCE), Abstract ID# : 382.

https://www.linkedin.com/posts/alex-dahoue-b7411536a_certificate-activity-7405674191913652224-0BsV?utm_source=share&utm_medium=member_desktop&rcm=ACoAAft_ZUGBtQ-1PtrzQAvTY0laB0NQ2GM0I4

Modeling and Artificial Intelligence for Ecological Prediction : This is a theoretical model that combines differential equations and neural networks to predict the dynamics of prey-predator populations; it was accepted on February 16, 2025, for the conference “INTERNATIONAL CONFERENCE ON ADVANCED SUSTAINABILITY ENGINEERING AND TECHNOLOGY 2025” after evaluation. [Mes projets de recherche/Research work_IA.pdf at main · Alex542277/Mes projets de recherche](#)

Scholarships, Awards, and Achievements :

2026 : **Travel grant.** Deep Learning Indaba

2026 : USAII (United States Artificial Intelligence Institute) **Global AI Hackathon**

[My-Piece/Certificate Global Hackathon AI.jpeg at main · Alex542277/My-Piece](#)

2025 : **Special Mention.** International Youth Math Challenge (IYMC), Qualification le 15 Décembre : https://www.linkedin.com/posts/alex-dahoue-b7411536a_qualification-activity-7406429028741558272cLyE?utm_source=share&utm_medium=member_desktop&rcm=ACoAAft_ZUGBtQ1PtrzQAvTY0laB0NQ2-GM0I4

2025 : **Award Certificate** for the presentation at the Second Pan-African University Conference :

https://www.linkedin.com/posts/alex-dahoue-b7411536a_certificate-activity-7405674191913652224-0BsV?utm_source=share&utm_medium=member_desktop&rcm=ACoAAft_ZUGBtQ1PtrzQAvTY0laB0NQ2-GM0I4

2025 : **Recherche Fellowship CIMPA** (International Center for Pure and Applied Mathematics) on modern mathematical tools in mechanics, September 8–19

https://www.linkedin.com/posts/alex-dahoue-b7411536a_attestation-activity-7409681420140310528-98zy?utm_source=share&utm_medium=member_desktop&rcm=ACoAAft_ZUGBtQ-1PtrzQAvTY0laB0NQ2GM0I4

2025 : **Finalist**. MIABE Hackathon. 28 juillet. **DAROLLO TECHNOLOGIES CORPORATION** [My-Piece/certificat_mbh_edition3_DAHOUÉ Labi Alexandre.pdf at main · Alex542277/My-Piece](#)

2023 : **2nd Prize in the OLYMPIA MATHS Competition** organized by the Federation of Students for Renewal and Total Success (FERST) at the National University of Science, Technology, Engineering, and Mathematics in Abomey on Saturday, July 15.

2021 : **National Scholarship for Excellence** awarded by the Beninese government to the top students following the national entrance exam for the National Higher Institute of Preparatory Classes for Engineering Studies, until they earn their degree in design engineering.

2021 : Class **Representative** and Class **Leader**.

2018 : Recipient of the **Valedictorian Award** from the NGO ADE (Action for Endogenous Development) and Prevention Ambassador for the « Zero School Pregnancies » program.

Presentations, workshops, and conferences :

2026 : **Workshop GeoAI**. Deep Learning Indaba

2026: **Poster presentation**. Titled “Hybrid AquaTwin-Drip Digital Twin for Drip Irrigation Management”, [Alex542277/AquaTwin-Drip-Hybrid-Digital-Twin-for-Controlling-Drip-Irrigation: The Challenge of Irrigation: From Empirical Decision-Making to Precision Management](#).

2026 : **Workshop**. Mathematical models for geophysical flows. IHP (Institut Henri Poincaré)

2026 : **QBronze 186 Workshop**. Quantum Computing and Programming Workshop organised by QWorld

2025 : **Participation and Presentation**. Second International Conference of the Pan-African University

2025 : **Participant**. Lectures and presentations of research work during the CIMPA school on modern mathematical tools in mechanics, September 8–19 [Post](#) | [LinkedIn](#)

2024 : **Participant**. Introduction to Data Science and Web Programming at the University of Lomé at CERSA (Regional Center of Excellence in Avian Sciences) from December 16 to 20.

Relevant Courses :

Udemy : Gaussian Process Regressor

-Supervised learning with the Gaussian process, Kernels and their usefulness, Scikit-learn Gaussian Process Regressor

Udemy : Physics Informed Neural Networks (PINNs). [My-Piece/Certificate_PINNs_and_I-PINNs.pdf at main · Alex542277/My-Piece](#)

-Pytorch basis, Finite Difference Method Numerical Solution 1D Heat Equation, Finite Difference Method Numerical Solution for 2D Burgers Equation

-PINNs solution for 1D Burgers Equation, PINNs solution for 2D Heat Equation, Deepxde solution for 1D Heat, Deepxde solution for 2D Navier Stokes

Udemy : Introduction to AI and Quantum Computing

[My-Piece/certificat_Quantum_Computing_Alexandre_Dahoue.pdf at main · Alex542277/My-Piece](#)

-Introduction to AI and Quantum Computing

-In-Depth Exploration of Artificial Intelligence

-Quantum Computing (Quantum Computing for Beginners, Quantum Circuits and Gates, Quantum Algorithms)

-Quantum Data Processing for AI, Quantum Neural Networks

-Optimization Techniques for Quantum AI

CIMPA SCHOOL 2025: [My-Piece/CIMPA courses Dahoue.pdf at main · Alex542277/My-Piece](#)

- Numerical Solution of Partial Differential Equations (Finite Volume Method in Discrete Duality) for Anisotropic Flow Problems.
- Numerical Methods and Scientific Computing
- Analytical and Numerical Methods for Differential Equation Models
- Mathematical modeling and numerical simulation of complex phenomena described by ordinary differential equations and partial differential equations.
- Machine Learning and Deep Learning for Engineers

National Higher School of Mathematical Engineering and Modeling

- Numerical modeling, Optimization and stochastic simulation (white noise, Wiener process or Brownian motion, Fokker-Planck equation, constrained optimization in probability, Itô and Stratonovich stochastic integrals)
- Numerical Analysis: Finite-precision arithmetic, interpolation (linear, Newton, Hermite, cubic, spline), Legendre polynomials, iterative methods (Jacobi, Gauss-Seidel, Krylov approximation technique), numerical methods for solving partial differential equations (finite difference method, finite element method, and finite volume method)
- Deterministic and Probabilistic Numerical Projects (Numerical solution of ordinary differential equations, Numerical solution of partial differential equations using numerical methods, Monte Carlo methods for probabilistic problems, Monte Carlo simulation, simulation of continuous and discrete probability distributions using various methods, simulation of Markov chains)
- Nonstationary Problems (Cauchy Problems, Time-Dependent Boundary Value Problems)
- Operations Research (linear and nonlinear programming, genetic algorithms, simulated annealing, bee colony optimization) Continuous and discrete Markov chains
- Inferential statistics (parametric estimation (maximum likelihood method, method of moments, confidence interval), hypothesis testing), Modeling Random Phenomena
- Imperative programming, advanced programming (parallel architectures, parallel computing, distributed computing)
- Analysis and Database, Information Transmission, Signal Processing, Imaging and Pattern Recognition

Career history :

<u>Date</u>	<u>Position</u>	<u>Institution</u>
2025	Project manager	INNOV-GROUP
2025	President	Mathematical Modeling Club
2025	Secretary-General	Federal Entity Section Office
2023	Web Developement Internship	Pro Technology Plus
2023-Present	Deputy Secretary General	Association of Students from Mono-Couffo

Technical Skills

Python, Sklearn, Matlab, R, C/C++,

Fortran, SQL, Jupiter Notebook,
Visual Studio Code, Pytorch, TensorFlow

Soft skills

Critical thinking, organizational skills, and attention to detail

Teamwork and Group Work, Communication,
Time Management, The Art

Research Interest

Foundational AI, AI for Science, Data science
robotics, digital twins

Area of Interest

Reading, football, jogging, games, guitar

Languages

English, French

References

Available upon request